AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

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THE SITE

Location	Meta, OR -- station KDLS, America/Los_Angeles
Committed pair	OSM 321385147 -> 373945186 Meta / Meta
Facade gap	140.6 m between the two halls
Weather record	42,451 real hourly records from KDLS
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3493 C at 275 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-17 (clear_cool)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	none
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 13 of 24 hours with 2 mode changes. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	13 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	5.574	18.0	4.441	1.280
01	FREE	yes	-	6.264	18.0	5.243	1.395
02	FREE	yes	-	6.264	18.0	5.243	1.337
03	FREE	yes	-	5.652	18.0	4.443	1.168
04	FREE	yes	-	7.343	18.0	6.115	1.433
05	FREE	yes	-	7.000	18.0	6.156	1.260
06	FREE	yes	-	7.411	18.0	6.440	1.314
07	FREE	yes	-	10.347	18.0	9.686	1.374

08	FREE	yes	-	10.856	18.0	7.780	1.836
09	FREE	yes	-	13.827	18.0	11.356	2.176
10	FREE	yes	-	16.299	18.0	12.990	2.081
11	FREE	yes	-	16.071	18.0	14.100	2.079
12	FREE	yes	-	17.646	18.0	14.171	2.166
13	mechanical	no	dry-bulb	18.514	18.0	13.925	2.240
14	mechanical	yes	-	17.628	18.0	12.226	2.124
15	mechanical	yes	-	16.142	18.0	11.145	1.898
16	mechanical	yes	switch budget	14.599	18.0	10.124	1.738
17	mechanical	yes	switch budget	12.855	18.0	9.483	1.572
18	mechanical	yes	switch budget	10.997	18.0	8.925	1.288
19	mechanical	yes	switch budget	10.172	18.0	8.343	1.666
20	mechanical	yes	switch budget	9.226	18.0	7.360	1.678
21	mechanical	yes	switch budget	9.271	18.0	7.233	1.863
22	mechanical	yes	switch budget	9.055	18.0	7.229	1.661
23	mechanical	yes	switch budget	8.528	18.0	7.263	1.497

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.573 C, which is 12.427 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.280 C, and the margin is measured, not chosen: 0.986 C of group-conditional forecast error for this hour of day, plus 0.1419 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.264 C, which is 11.736 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.395 C, and the margin is measured, not chosen: 0.934 C of group-conditional forecast error for this hour of day, plus 0.3091 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.264 C, which is 11.736 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.337 C, and the margin is measured, not chosen: 0.876 C of group-conditional forecast error for this hour of day, plus 0.3091 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.653 C, which is 12.347 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.168 C, and the margin is measured, not chosen: 0.795 C of group-conditional forecast error for this hour of day, plus 0.2209 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.343 C, which is 10.657 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.432 C, and the margin is measured, not chosen: 1.031 C of group-conditional forecast error for this hour of day, plus 0.2495 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.000 C, which is 11.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.259 C, and the margin is measured, not chosen: 0.929 C of group-conditional forecast error for this hour of day, plus 0.1781 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.411 C, which is 10.589 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.314 C, and the margin is measured, not chosen: 0.888 C of group-conditional forecast error for this hour of day, plus 0.2740 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.347 C, which is 7.653 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.374 C, and the margin is measured, not chosen: 1.032 C of group-conditional forecast error for this hour of day, plus 0.1905 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.856 C, which is 7.144 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.836 C, and the margin is measured, not chosen: 1.545 C of group-conditional forecast error for this hour of day, plus 0.1391 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.827 C, which is 4.173 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.176 C, and the margin is measured, not chosen: 1.734 C of group-conditional forecast error for this hour of day, plus 0.2903 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.299 C, which is 1.701 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.081 C, and the margin is measured, not chosen: 1.680 C of group-conditional forecast error for this hour of day, plus 0.2494 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.071 C, which is 1.929 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.080 C, and the margin is measured, not chosen: 1.673 C of group-conditional forecast error for this hour of day, plus 0.2545 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.646 C, which is 0.354 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.166 C, and the margin is measured, not chosen: 1.780 C of group-conditional forecast error for this hour of day, plus 0.2338 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.514 C against a 18.0 C limit, so it fails by 0.514 C. It would take a limit 0.514 C higher, or a bound 0.514 C tighter, to change this hour.

14:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.628 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.143 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.599 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.855 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.997 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.172 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.226 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.272 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.055 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.528 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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